7.3 What Makes a Determinant

WORKSHEET

The question

The domain of the function f is the set of all 2×2 matrices and its range is the set of real numbers. Thus, if $\mathbf{M}$ is a 2×2 matrix, then $\mathrm{f}(\mathbf{M}) \in \mathbb{R}$.

The function f has the property that $\mathrm{f}(\mathbf{MN}) = \mathrm{f}(\mathbf{M})\mathrm{f}(\mathbf{N})$ for any 2×2 matrices $\mathbf{M}$ and $\mathbf{N}$.

(i) You are given that there is a matrix $\mathbf{M}$ such that $\mathrm{f}(\mathbf{M}) \neq 0$. Let $\mathbf{I}$ be the 2×2 identity matrix. By considering $\mathrm{f}(\mathbf{MI})$, show that $\mathrm{f}(\mathbf{I}) = 1$.

(ii) Let $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. You are given that $\mathrm{f}(\mathbf{J}) \neq 1$. By considering $\mathbf{J}^2$, evaluate $\mathrm{f}(\mathbf{J})$.

Using $\mathbf{J}$, show that, for any real numbers a, b, c and d,

$$\mathrm{f}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = -\mathrm{f}\begin{pmatrix} c & d \\ a & b \end{pmatrix} = \mathrm{f}\begin{pmatrix} d & c \\ b & a \end{pmatrix}.$$

(iii) Let $\mathbf{K} = \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$ where $k \in \mathbb{R}$. Use $\mathbf{K}$ to show that, if the second row of the matrix $\mathbf{A}$ is a multiple of the first row, then $\mathrm{f}(\mathbf{A}) = 0$.

(iv) Let $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. By considering the matrices $\mathbf{P}^2$, $\mathbf{P}^{-1}$, and $\mathbf{K}^{-1}\mathbf{PK}$ for suitable values of k, evaluate $\mathrm{f}(\mathbf{P})$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What does each special matrix in the question do? Work out $\mathbf{JM}$, $\mathbf{MJ}$, $\mathbf{KM}$ and $\mathbf{PM}$ for a general $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

STAGE 3 Dependent rows

Ask yourself First consider a matrix $\mathbf{B}$ with two *equal* rows. What is $\mathbf{JB}$? What does that say about $\mathrm{f}(\mathbf{B})$?

STAGE 4 The shear

Pause and try Compute $\mathbf{P}^2$ and $\mathbf{K}^{-1}\mathbf{PK}$. Choose k to make them equal. What does that give for $\mathrm{f}(\mathbf{P})$?

Try this next

1. **Other examples.** Show that $\mathrm{f}(\mathbf{M}) = (\det \mathbf{M})^3$ satisfies all the conditions of the question, and that $\mathrm{f}(\mathbf{M}) = |\det \mathbf{M}|$ satisfies the multiplicative property but not $\mathrm{f}(\mathbf{J}) \neq 1$.

2. **Column operations.** Show that f is also zero on any matrix whose second column is a multiple of the first.

3. **Scalings.** Let $\mathbf{K}_k = \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$. Show that $\mathrm{f}\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix} = \mathrm{f}(\mathbf{K}_k)$, and that $\mathrm{f}(\mathbf{K}_k)\mathrm{f}(\mathbf{K}_l) = \mathrm{f}(\mathbf{K}_{kl})$.

4. **Reduction to the determinant.** HARDERShow that every invertible 2×2 matrix can be written as a product of matrices of the forms $\mathbf{J}$, $\mathbf{K}_k$ and $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$. Deduce that $\mathrm{f}(\mathbf{M})$ depends only on $\det \mathbf{M}$.

Further reading

- **Sheldon Axler, *Linear Algebra Done Right*** (Springer; free online). Defines the determinant from its properties, in the spirit of this question.
- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Chapter 5 devel-

ops determinants from three rules.

- **3Blue1Brown, "The determinant"** (video, free online). The area interpretation, beautifully animated.

The full guided solution, and what the question leads to at university, is in Section 7.3:
https://beyondstep.keithmcnulty.org/ch07/determinant-axioms.html